

THE SILENCE EXPERIMENT

A Multi-State Protocol for Detecting Consciousness-Dependent
Decoherence in Shielded Environments

THEORETICAL PHYSICS • CONSCIOUSNESS STUDIES • QUANTUM FOUNDATIONS

By

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*Companion paper: The Silence Paradigm — A Theoretical Framework
for Investigating Consciousness-Dependent Quantum Decoherence*

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SECTION I

Abstract

This paper proposes a controlled experimental protocol for testing the von Neumann–Wigner (VNW) interpretation of quantum mechanics—the hypothesis that consciousness plays a role in quantum state reduction—against the prevailing environmental decoherence framework (Zurek, Zeh, Joos). Building on prior consciousness-quantum interaction studies with enhanced controls and instrumentation, the protocol measures whether the cognitive state of a nearby conscious observer measurably affects the decoherence rate of a prepared quantum system under controlled conditions.

The experimental platform places a single experienced meditator in states showing EEG patterns consistent with objectless awareness inside a shielded environment. A quantum coherence sensor measures decoherence rates across seven controlled conditions: three controls (empty chamber, non-meditator resting, active cognitive task) and four meditation states (focused attention, open monitoring, objectless awareness/cessation, and non-directed openness). This multi-state design—addressing a key methodological gap in prior research where the relationship between specific meditation styles and observer effects was not empirically established—allows data to determine which consciousness state, if any, produces the largest effect.

A secondary experiment extends the Princeton Engineering Anomalies Research (PEAR) lineage using quantum random number generators with improved shielding, blinding, and instrumentation. Exploratory biophotonic monitoring provides an additional independent biomarker.

The protocol is phased: a Micro-Pilot (~\$2,000–\$5,000) using commercial QRNGs; a Phase 0 proof-of-concept (~€70,000–€150,000) using nitrogen-vacancy centers in diamond; and a Phase 1 full protocol (~€500K–€2M) using superconducting qubits in an underground laboratory. The protocol employs triple-blind analysis and pre-registered statistical thresholds. Phase 0 is explicitly acknowledged as underpowered for very small effects; it can detect medium-to-large effects ($d > 0.3$) but not the very small effects ($d \approx 0.0002$) reported in prior literature.

This protocol is proposed as a contribution to an ongoing line of experimental inquiry into consciousness-quantum interactions, not as a definitive resolution. Its value lies in the specificity of its predictions and the rigor of its controls. Null results establish upper bounds; positive results require replication.

SECTION II

Introduction: The Measurement Problem and the Observer

Quantum mechanics contains an unresolved foundational question known as the measurement problem. A quantum system exists in superposition—multiple states simultaneously—until it undergoes a transition to a single definite outcome. What causes this transition has been debated since the formalization of quantum theory in the late 1920s and remains unresolved today.

Every major interpretation of quantum mechanics is, at its core, a different answer to this question. The Copenhagen interpretation treats measurement as a primitive concept without further analysis. The Many-Worlds interpretation (Everett, 1957) denies that collapse occurs at all. Bohmian mechanics introduces hidden variables. QBism treats quantum states as subjective degrees of belief. And the von Neumann–Wigner (VNW) interpretation, first articulated by von Neumann in 1932 and extended by Wigner in 1961, proposes that consciousness itself is the agent of state reduction.

The VNW interpretation is not mainstream. It is, however, not empirically refuted. Its marginal position in contemporary physics reflects philosophical discomfort, the success of environmental decoherence theory, and the absence of controlled experiments designed to distinguish between interpretations using the observer's cognitive state as an experimental variable.

2.1 Environmental Decoherence: The Mainstream Position

The environmental decoherence program (Zurek, 2003; Zeh, 1970; Joos et al., 2003) provides a mechanism by which quantum superpositions become effectively classical through interaction with the environment. Decoherence explains why macroscopic objects do not exhibit quantum interference, why certain bases are preferred in measurement (einselection), and why the classical world appears stable. It is the dominant framework in contemporary quantum foundations.

However, decoherence does not solve the measurement problem in its entirety. As Zurek has acknowledged, decoherence explains the suppression of interference but does not explain why a specific outcome is realized from among the surviving diagonal elements. This residual gap—the problem of definite outcomes—is where the VNW interpretation operates.

2.2 The Boundary Effect

The theoretical framework (Paper 1: The Silence Paradigm) proposes a 'boundary effect': decoherence is the quantum-classical boundary. If consciousness participates in state reduction, the effect should manifest at this boundary as a shift in coherence times. This experiment measures not consciousness itself but the shift in the decoherence boundary associated with the observer's state.

2.3 The Experimental Gap

Controlled experiments systematically varying the cognitive state of a conscious observer while measuring the decoherence rate of a nearby quantum system represent a gap in the existing literature. This protocol aims to

address that gap, building on prior work by Radin et al. (2012, 2016) and Jahn and Dunne (2005) with improved methodology, more precise state characterization, and a decoherence-focused measurement strategy.

SECTION III

Theoretical Context

3.1 The Von Neumann–Wigner Interpretation

In his 1932 treatise *Mathematical Foundations of Quantum Mechanics*, von Neumann demonstrated that the boundary between the quantum system and the measuring apparatus can be moved arbitrarily far along the measurement chain without affecting the predictions of the theory. The logical terminus of this chain is the conscious observer. Wigner (1961) made the stronger claim explicit: consciousness is necessary for quantum state reduction. Stapp (2007) developed this view further.

The present experiment tests a weaker, empirically tractable version: if consciousness participates in state reduction, then varying the cognitive state of a nearby conscious observer should produce measurable differences in decoherence rates.

3.2 Environmental Decoherence Theory

Environmental decoherence theory predicts that the cognitive state of a nearby conscious observer should have no effect on the decoherence rate of a quantum system, provided environmental coupling remains constant. This is the null prediction the experiment is primarily designed to test.

3.3 Orchestrated Objective Reduction (Orch-OR)

Penrose and Hameroff (2014) proposed that consciousness arises from quantum processes in neuronal microtubules, with gravity triggering collapse at a threshold. Orch-OR makes distinct predictions from VNW, though quantitative predictions for this specific experimental configuration have not been derived. This is a limitation.

3.4 Neuroscientific Basis for Meditation as a Controlled Variable

Deep meditation produces neurologically distinctive states well-documented in the peer-reviewed literature. Lutz et al. (2004) reported elevated gamma-band power and long-range phase synchrony in long-term meditators. Berkovich-Ohana et al. (2012) documented EEG correlates of mindfulness-induced changes in gamma band activity, including DMN deactivation. Winter et al. (2020) provided further characterization of neural markers associated with content-free awareness states.

Key features include: gamma wave coherence, default mode network suppression (Brewer et al., 2011), distinct differentiation from sleep, and structural brain changes. EEG identifies neural correlates consistent with these states, not the subjective states themselves. EEG provides convergent evidence for the target state, not definitive verification of subjective experience.

3.5 Speculative Theoretical Context

Several additional frameworks are relevant as speculative context: the Wheeler-DeWitt equation (timelessness in canonical quantum gravity), the ER=EPR conjecture (Maldacena & Susskind, 2013), Integrated Information Theory (Tononi, 2004), and Wheeler's Participatory Anthropic Principle. None are confirmed, and none are relied upon for the design or predictions of this experiment.

SECTION IV

The Mechanism Problem

No confirmed physical mechanism currently exists by which the cognitive state of a conscious observer would alter the decoherence rate of a nearby quantum system beyond the known effects of environmental decoherence. The physical coupling—if it exists—is not specified by any current theory at a level that yields quantitative predictions for this experimental configuration.

This limitation is stated directly because it is important. The VNW interpretation posits a role for consciousness in state reduction but does not specify the mechanism. Orch-OR proposes a specific mechanism but has not derived quantitative predictions for the effect of a meditator on a separate quantum system. No existing theory provides a calculation of the expected effect size.

The experiment is nevertheless worth conducting for five reasons:

- **First:** The measurement problem is genuinely unresolved. Environmental decoherence explains the suppression of interference but not the realization of definite outcomes.
- **Second:** Controlled experiments systematically varying consciousness state while measuring decoherence represent a gap in the existing literature that this protocol aims to address.
- **Third:** Science regularly tests hypotheses before mechanisms are fully understood. John Snow identified cholera transmission decades before germ theory.
- **Fourth:** The cost is low relative to the significance of the question.
- **Fifth:** Null results are valuable, establishing upper bounds on consciousness-dependent quantum state reduction and constraining the parameter space of consciousness-inclusive interpretations.

SECTION V

Experimental Design

5.1 Multi-State Consciousness Conditions

The multi-state design addresses a key methodological critique of prior consciousness-quantum research: the relationship between specific meditation styles and observer effects is not empirically established. This design allows the data to determine which state—if any—produces the effect, rather than assuming the answer in advance.

The protocol specifies seven controlled conditions:

Condition	Description	EEG Markers	Rationale
Control 1: Empty Chamber	No human present	N/A	True baseline — eliminates all human variables
Control 2: Non-Meditator Resting	Untrained person sits in chamber, eyes closed, relaxed	Normal resting-state EEG (alpha dominant)	Controls for mere human presence, body heat, EM from nervous system
Control 3: Active Cognitive Task	Meditator performs mental arithmetic or verbal tasks	High beta, active frontal regions	Controls for cognitive engagement — the “noisy mind” condition
Condition A: Focused Attention (Shamatha)	Experienced meditator in concentrated focus on single object (breath)	Increased beta/gamma coherence, frontal midline theta	Tests whether concentrated attention affects decoherence
Condition B: Open Monitoring (Vipassana)	Experienced meditator in receptive awareness, no specific focus	Increased theta, decreased beta, parietal changes	Tests whether open receptive awareness affects decoherence
Condition C: Objectless Awareness / Cessation	Experienced meditator in contentless awareness or full cessation (nirodha)	Gamma burst followed by EEG flattening, or sustained high-amplitude gamma	Tests the primary hypothesis — maximum cognitive silence
Condition D: Non-Directed Openness	Experienced contemplative practitioner in complete receptive surrender without specific technique	Variable — possibly high alpha coherence, heart-brain synchronization	Tests whether non-technique-based receptive openness produces different effects

Table 1: The seven controlled consciousness conditions.

A note on Condition D: The theoretical framework suggests the mechanism may involve receptive openness rather than technique-based cognitive reduction. If the effect is mediated through receptive openness rather than through progressive silencing, Condition D might produce different results than Condition C. The multi-state design allows this question to be answered empirically.

5.2 The Shielded Chamber and Five-Layer Isolation

The chamber minimizes all known sources of decoherence so that any residual variation across conditions cannot be attributed to uncontrolled environmental factors. The chamber is a perfect $3.0\text{ m} \times 3.0\text{ m} \times 3.0\text{ m}$ cube with a dual-door airlock entry system on the east face (see TSP-ENG-001 Rev B for full engineering specifications). Five isolation layers address specific confounds:

Layer	Technology	Purpose
1. Acoustic + Thermal	Closed-cell acoustic foam + aerogel blanket (150 mm)	Eliminate airborne noise and thermal drift ($\pm 0.01\text{ K}$)
2. Magnetic Shielding	Mu-metal enclosure (3 mm, hydrogen-annealed)	Attenuate DC and low-frequency magnetic fields ($>60\text{ dB}$)
3. RF/Microwave Shielding	OFHC copper sheet (2 mm), welded seams	Eliminate external EM radiation ($>100\text{ dB}$, 10 kHz – 10 GHz)
4. Vibration Isolation	Pneumatic isolators + neoprene dampers (200 mm)	Sub-nanometer vibration control ($<10^{-6}\text{ m/s}^2$)
5. Structural / Faraday	Aluminum shell (5 mm) + copper mesh cage	Full EM isolation ($>120\text{ dB}$ combined) and mechanical rigidity

Table 2: Five-layer isolation specifications. Each layer addresses a specific confound.

For Phase 1, deep underground location ($\geq 1,000\text{ m}$ rock overburden) provides cosmic ray attenuation $\geq 10^6$. Recommended facilities: SURF (Lead, SD), SNOLAB (Sudbury, ON), LNGS (Gran Sasso, Italy), or Boulby (North Yorkshire, UK).

5.3 Quantum Sensor Options

Parameter	Option A: Superconducting Transmon Qubit	Option B: NV-Center in Diamond
T_2 (decoherence time)	50–200 microseconds	1–10 milliseconds at room temperature
Operating temperature	$\sim 15\text{ mK}$ (dilution refrigerator)	Room temperature
Cost	$\sim \$500\text{K}$ – $\$1\text{M}$ (cryogenic system)	$\sim \$50\text{K}$ – $\$100\text{K}$
Sensitivity	Highest sensitivity to environmental perturbation	Lower sensitivity, longer baseline T_2
Advantages	Maximum detection sensitivity	Cheaper, simpler, no cryogenics
Disadvantages	Expensive, complex, requires specialized operators	Less sensitive to small perturbations

Table 3: Quantum sensor options for decoherence measurement.

Recommendation: NV-center in diamond for Phase 0 (proof of concept). If a positive signal is detected, upgrade to superconducting qubit for Phase 1. This phased approach minimizes initial cost while preserving sensitivity for definitive testing.

5.4 Instrument Array and Environmental Monitoring

Instrument	Measures	Target	Sensitivity
Quantum Coherence Sensor (NV or transmon)	Decoherence rates (T_2)	Primary experiment	$\leq 1 \mu\text{s}$ resolution
QRNGs (×4)	Quantum randomness statistics	Secondary experiment	≥ 1 Mbit/s, quantum sourced
Single-Photon Detectors	Biophotonic emissions	Exploratory	< 100 dark counts/s
64-Channel Wireless EEG	Neural state characterization	Independent variable	≥ 1 kHz sampling
SQUID Magnetometers	Magnetic fields (femtotesla)	Environmental monitor	< 10 fT/√Hz
Temperature Sensors (×4)	Thermal environment inside shield	Artifact control	$\pm 0.001^\circ\text{C}$ resolution
3-Axis Fluxgate Magnetometer	Ambient magnetic field inside shield	Artifact control	Continuous recording
3-Axis Accelerometer	Vibration inside chamber	Artifact control	10^{-9} g, continuous

Table 4: Instrument array. All environmental channels recorded synchronously with quantum sensor data and EEG via GPS-disciplined 10 MHz common time base ($< 1 \mu\text{s}$ inter-channel synchronization).

If any anomaly appears in quantum data, environmental channels confirm whether it correlates with a physical artifact or is specific to the consciousness condition. This is the most critical step in the artifact analysis.

5.5 Session Protocol (Revised)

The session protocol has been expanded to include detailed pre-session preparation, contamination control, airlock procedures, and post-session grounding. All sessions follow a strict single-observer rule.

5.5.1 Pre-Session Preparation (24–48 Hour Window)

To minimize residual cognitive and physiological excitation, each observer shall undergo a standardized 24–48 hour pre-session preparation protocol. This includes:

- Controlled low-excitation diet: no caffeine, alcohol, or high-glycemic foods
- Complete avoidance of electronic devices and metallic objects
- Minimum 8-hour sleep period immediately prior to chamber entry

The goal is to establish a stable baseline of minimum systemic excitation before the observer enters the airlock.

5.5.2 Garment and Contamination Control

Observers shall wear only facility-provided non-metallic, low-electromagnetic-signature garments (100% cotton or synthetic-free linen equivalent). No jewelry, conductive materials, synthetic fibers, or external devices are permitted inside the chamber. All personal items are removed and stored in the external preparation area prior to airlock entry.

5.5.3 Airlock Stabilization and Buffer Phase

Upon entering the airlock, the observer remains for a 60-second stabilization period. During this interval:

- RF power meter confirms residual electromagnetic emission is below -80 dBm (10 kHz–10 GHz)
- Fluxgate magnetometer confirms no ferromagnetic contamination
- Controlled HEPA-filtered air stream (particle density <10 cm⁻³) may be applied for 5 minutes to establish a uniform low-turbulence buffer medium

Only after contamination check passes is the inner door released and the observer permitted to enter the main chamber.

5.5.4 Strict Single-Observer Rule

Only one observer is permitted inside the main cubic volume during any active data-collection session. No assistants, additional personnel, or equipment beyond the pre-installed sensor array shall be present. This isolates the independent variable (the single observer's cognitive state) and eliminates inter-observer coupling or social facilitation effects.

5.5.5 Session Timing and Duration

Each session follows a standardized procedure:

- **Step 1:** 30-minute equipment calibration and baseline (quantum sensor running, chamber empty)
- **Step 2:** Observer enters via airlock, connected to EEG. 10-minute settling period (data discarded)

- **Step 3:** 45-minute condition period (meditation, resting, or cognitive task)
- **Step 4:** Real-time EEG monitoring confirms neural patterns consistent with the target state. Sessions without consistent markers are flagged and analyzed separately
- **Step 5:** 15-minute post-session baseline (observer exits, quantum sensor continues)
- **Total:** ~100 minutes per session

Each observer session is limited to a single 70-minute active window (45 minutes condition + 25 minutes baseline) within any 24-hour period. This constraint maintains minimum cumulative excitation and ensures statistical independence between sessions.

5.5.6 Post-Session Exit and Grounding Protocol

After completion of the active session, the observer remains in the airlock for a mandatory 10-minute post-session grounding period with continuous environmental monitoring. Only after this period is the outer door released and the observer permitted to return to normal laboratory conditions. This step prevents abrupt re-exposure to external excitation sources and allows full physiological and electromagnetic re-equilibration.

5.5.7 Emergency and Override Procedures

In the event of any safety threshold breach ($O_2 < 19.5\%$, excessive vibration, or observer distress), the inner-door panic bar allows immediate egress without tools. All sessions are monitored externally via fiber-optic video and intercom. At no point during normal operation are both airlock doors open simultaneously.

5.6 Participants

- Conditions A, B, C: Minimum 5 experienced meditators with 10,000+ hours documented practice and demonstrated ability to produce EEG patterns consistent with target states
- Condition D: Minimum 3 experienced contemplative practitioners with documented practice in non-directed receptive openness
- Control 2: Minimum 5 non-meditators matched for age and sex
- Control 3: Same meditators as Conditions A–C (within-subjects comparison)
- Sessions: Each participant completes minimum 50 sessions per condition, randomized order

5.7 Blinding and Replication

Triple-blind analysis: the subject does not know which instruments are active; instrument operators do not know the condition schedule; initial data analysts identify anomalous periods independently, which are then correlated with session logs by a separate team. Subject blinding is not possible for consciousness-state experiments—a meditator knows when they are meditating. This limitation is inherent to the design and is acknowledged explicitly.

Full replication: protocol repeated with different meditators, different instruments of the same type, and different facilities.

SECTION VI

Primary Experiment: Quantum Decoherence Test

The quantum decoherence test is designated as the primary experiment. It has the most specific predicted outcomes and the highest capacity to discriminate between competing interpretations of quantum mechanics.

6.1 Logic

If consciousness participates in wavefunction collapse (VNW), then the cognitive state of a nearby conscious observer should affect decoherence rates. The predicted VNW ordering: $T_{\text{coherence}}(\text{cessation}) > T_{\text{coherence}}(\text{baseline}) > T_{\text{coherence}}(\text{active cognition})$. The multi-state design allows testing of which meditation state, if any, produces the largest effect.

6.2 Theory Discrimination

Theory	Predicted Pattern	Key Discriminator
Von Neumann–Wigner	Cessation > Baseline > Cognition	Consciousness state affects decoherence rates
Environmental Decoherence	All conditions equal	No variation across any conditions
Orch-OR	Meditation $\neq$ Cognition (pattern TBD)	Consciousness affects local spacetime geometry
Epiphenomenalism	All conditions equal	Identical to decoherence prediction

Table 5: Adversarial theory-testing matrix.

6.3 Full Adversarial Prediction Matrix

Theory	Decoherence Test	QRNG Test	Biophotonics
Von Neumann–Wigner	Cessation > Baseline > Cognition	Deviation during meditation	No specific prediction
Environmental Decoherence	No difference across conditions	No deviation	No specific prediction
Orch-OR	Meditation $\neq$ Cognition	Possible deviation	Possible biophotonic changes
Epiphenomenalism	No difference across conditions	No deviation	No specific prediction

Table 6: Full adversarial prediction matrix across experimental channels.

SECTION VII

Secondary Experiment: Quantum Random Number Generator Test

The Princeton Engineering Anomalies Research (PEAR) laboratory (Jahn & Dunne, 2005) operated from 1979 to 2007, conducting millions of trials on consciousness–random number generator interaction. PEAR reported small but statistically significant deviations from randomness ($d \approx 0.0002$). The Global Consciousness Project (Nelson, 2001, 2015) extended this with a worldwide RNG network.

It should be noted that PEAR’s effects were not independently replicated to the satisfaction of the broader scientific community, and a meta-analysis (Bösch, Steinkamp, & Boller, 2006) found effect sizes correlated inversely with methodological quality. These are significant criticisms that the present protocol is specifically designed to address through underground shielding, four independent quantum-source generators ($\geq 10^9$ bits per session), triple-blind analysis, and EEG-characterized consciousness states rather than diffuse ‘intention’ or ambient collective attention.

The QRNG test is independent of the decoherence test. A positive result on one with a null result on the other would carry distinct theoretical implications.

SECTION VIII

Exploratory: Biophotonic Monitoring

Research by Fritz-Albert Popp and others documented that living cells emit ultra-weak photon emissions (biophotons) correlating with biological state. Single-photon detectors in the dark chamber provide an opportunity to monitor whether deep meditation alters biophotonic emission patterns. Biophotonic monitoring is classified as exploratory. No specific predictions are pre-registered. It provides an additional independent biomarker and maximizes the scientific return of the platform.

SECTION IX

Data Analysis and Statistical Framework

9.1 Pre-Registration

Complete analysis plan pre-registered on OSF or AsPredicted before any data collection. Includes: primary hypothesis, secondary hypotheses, planned sample size, statistical tests, stopping rules, correction for multiple comparisons, and criteria for conclusions.

9.2 Primary Analysis

- Compare mean T_2 across all 7 conditions using one-way repeated-measures ANOVA (or mixed-model ANOVA for between-subjects comparisons involving non-mediator controls)
- Post-hoc pairwise comparisons with Bonferroni correction
- Primary comparison: Condition C (cessation) vs. Control 1 (empty chamber)
- Secondary: all meditation conditions vs. all controls
- Effect size: Cohen's d for each comparison
- Significance threshold: $p < 0.001$ (conservative, given extraordinary nature of claim)
- Bayesian analysis in parallel: Bayes factors ($BF > 10$ = strong evidence, $BF > 100$ = decisive)

9.3 Secondary Analysis

- QRNG: compare entropy, sequential correlations, and mean deviation across conditions
- Cross-correlation: real-time EEG features with simultaneous T_2 measurements
- Dose-response: within meditation conditions, do sessions with deeper EEG markers show larger T_2 effects?

9.4 Artifact Analysis

For every significant result, examine ALL environmental channels for correlated changes. If T_2 correlates with ANY environmental variable (temperature, magnetic field, vibration), classify as artifact. This is the most critical step in the analysis pipeline. A result is only classified as a candidate consciousness effect if no environmental correlate is found.

SECTION X

Power Analysis

A conventional power analysis requires an expected effect size derived from theory or prior data. For the decoherence test, neither is available. For the QRNG test, the PEAR dataset provides a reference effect size of $d \approx 0.0002$, though this is contested.

Expected effects from existing literature are approximately $d = 0.01$ – 0.05 (very small). To detect $d = 0.02$ at $p < 0.001$ with power = 0.80 requires approximately 25,000+ trials per condition. Phase 0 cannot provide this many trials.

Phase 0 can detect medium-to-large effects ($d > 0.3$) but NOT the very small effects reported in existing literature. Phase 0 is therefore a pilot—it can detect unexpectedly large effects or establish that the effect is smaller than $d = 0.3$. A Phase 0 null result does NOT mean the effect doesn't exist—it means it is smaller than Phase 0 can detect.

This limitation is acknowledged directly. The experiment is exploratory in its effect-size expectations for the decoherence channel. Phase 1, with increased trial counts and more sensitive instrumentation, would be required to test for smaller effects.

SECTION XI

Stopping Rules

Criterion	Action
Phase 0: 500 total sessions minimum completed	Proceed to analysis
All $p > 0.10$	Stop. Publish null result with upper bounds.
Marginal signal ($0.001 < p < 0.05$)	Extend to 1,500 sessions for confirmation
Strong signal ($p < 0.001$ with clean environmental channels)	Proceed to Phase 1

Table 7: Pre-registered stopping rules.

Data collection continues for the pre-specified number of sessions regardless of interim results. No optional stopping.

SECTION XII

Budget, Phasing, and Timeline

12.1 Micro-Pilot (Pre-Phase 0)

Parameter	Specification
Budget	~\$2,000–\$5,000
Equipment	Commercial QRNG only (no quantum coherence measurement)
Location	Any quiet room with basic shielding
Participants	1–2 experienced meditators
Sessions	200+ per condition
Goal	Generate preliminary data, demonstrate ability to execute, attract collaborators. NOT a definitive test.

Table 8: Micro-Pilot specifications.

12.2 Phase 0 — Proof of Concept

Parameter	Specification
Budget	~€70,000–€150,000
Location	University lab with partial shielding
Quantum sensor	NV-center in diamond
EEG	Existing university system or rental
QRNG	Commercial unit
Shielding	Basic Faraday + mu-metal room
Personnel	PI + 1 postdoc + meditator participants (voluntary)
Timeline	12 months
Goal	Determine if ANY signal exists above noise floor. Go/no-go for Phase 1.

Table 9: Phase 0 specifications.

12.3 Phase 1 — Full Protocol

Parameter	Specification
Budget	~€500K–€2M
Location	Underground laboratory (SURF or SNOLAB)
Quantum sensor	Superconducting qubit system
Shielding	Full five-layer isolation with airlock

Parameter	Specification
Participant pool	Larger pool, extended data collection
Timeline	2–3 years
Goal	Definitive result with adequate statistical power

Table 10: Phase 1 specifications.

12.4 Cost Comparison

Experiment	Cost
Large Hadron Collider (CERN)	\$13.25B
James Webb Space Telescope	\$10.0B
LIGO	\$1.1B
LUX-ZEPLIN (LZ)	\$55M
Silence Experiment Phase 1	€500K–€2M
Silence Experiment Phase 0	€70K–€150K
Silence Experiment Micro-Pilot	\$2K–\$5K

Table 11: Cost comparison with major physics experiments.

12.5 Recommended Facilities

Facility	Location	Depth	Notes
SURF (Primary)	Lead, South Dakota	1,490 m (~4,300 m.w.e.)	Hosts LZ; US-based
SNOLAB	Sudbury, Ontario	2,070 m (~6,010 m.w.e.)	Deepest in N. America
LNGS	Gran Sasso, Italy	1,400 m (~3,800 m.w.e.)	Largest underground lab
Boulby	North Yorkshire, UK	1,100 m (~2,850 m.w.e.)	Hosts dark matter searches

Table 12: Recommended underground physics facilities (Phase 1).

12.6 Timeline

Phase 0: Months 1–3 facility negotiation and equipment procurement; Months 4–6 chamber construction, calibration, meditator screening; Months 7–11 data collection across conditions; Month 12 blinded analysis and manuscript. Phase 1 (if warranted): approximately 2–3 years from Phase 0 completion.

SECTION XIII

Ethical Considerations

IRB approval is required before data collection. The protocol monitors for claustrophobia, anxiety, and adverse meditation experiences (e.g., dissociation, distress). Informed consent includes disclosure of the speculative nature of the hypothesis, the possibility of null results, and the investigator's theoretical commitment to the framework.

Data anonymization procedures are specified. All data and analysis code will be deposited in an open repository upon publication. The investigator's theoretical commitment to the Silence Paradigm framework is disclosed as a potential source of bias, mitigated by the triple-blind analysis protocol and pre-registered statistical plan.

SECTION XIV

Limitations

The following limitations are stated explicitly:

- Phase 0 is underpowered for very small effects ($d < 0.3$). It functions as a pilot, not a definitive test.
- No established physical mechanism exists for consciousness-decoherence interaction.
- The multi-state design addresses but does not fully resolve the question of which meditation style is optimal.
- Underground location (Phase 1) introduces practical constraints on participant access and session scheduling.
- EEG-state correspondence is imperfect—neural markers are correlates, not confirmations of subjective state.
- Subject blinding is not possible for consciousness-state experiments.
- A positive result would establish correlation, not causation or mechanism.
- No quantitative effect-size prediction can be derived from current theory. The experiment is exploratory in its effect-size expectations.

SECTION XV

Future Directions

The following extensions are speculative and entirely contingent on positive results from the primary experiment: gravitational measurements (atom interferometry) during meditation states; vacuum energy density measurements (Casimir effect plates); precision time-correlation measurements; entangled photon pair monitoring; quantum noise analysis.

Future extensions may explore synchronized multi-observer configurations after the single-observer effect has been robustly established and replicated.

None of these extensions are proposed as part of this experiment.

SECTION XVI

Conclusion

This protocol extends prior consciousness-quantum interaction studies with enhanced controls, multi-state comparison, and direct quantum coherence measurement. The primary experiment measures decoherence rates across seven controlled consciousness conditions, testing the VNW interpretation against environmental decoherence theory and Orch-OR.

The limitations have been stated directly: no confirmed mechanism, no quantitative effect-size prediction, Phase 0 underpowered for very small effects. The protocol is publishable regardless of outcome. A null result establishes upper bounds on consciousness-dependent decoherence. A positive result opens a new domain of investigation requiring replication.

The theoretical basis for this protocol is detailed in the companion paper (Paper 1: The Silence Paradigm).

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